ISACSoil: Multi-Layer Soil Moisture Sensing with LoRa Cross-Soil Communication

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***Abstract*—This paper presents ISACSoil, a new integrated LoRa communication-sensing design that enables multi-layer soil moisture sensing by modeling fine-grained LoRa signal path loss during propagation. Different from existing LoRa-based soil moisture sensing, ISACSoil achieves accurate *multi-layer* soil moisture estimation over long distances with commercial-off-the-shelf (COTS) LoRa devices. The water content in soils brings more signal attenuation. However, in addition to soil moisture, polarization misalignment is another factor that can degrade signal power, making it challenging to model the relationship between signal power and soil moisture. To address this challenge, we first develop a method that uses a cross-soil communication configuration to mitigate the effects of polarization misalignment. Then, we develop a fine-grained model of signal attenuation, reflection, and refraction to characterize the power loss after multi-layer propagation. Finally, we formalize the multi-layer soil moisture estimation problem as an optimization problem. We develop a two-step method to search for the optimal soil moisture values across different layers, subject to three constraints: relative permittivity, transceiver locations, and received signal strength. We implement ISACSoil with COTS LoRa devices and evaluate it in a real soybean field. Our results demonstrate that ISACSoil reduces the Volumetric Water Content (VWC) error by 1.69%, corresponding to approximately a 40% improvement over the baselines.**

I. Introduction

LoRa is a widely adopted low-power wide-area network (LPWAN) technology, known for its low-power, long-range communication capabilities [1], and is a key enabler for rural Internet of Things (IoT) applications, particularly in precision agriculture [2]–[10]. For instance, soil moisture is a critical metric that reflects the water content in soil. Maintaining optimal soil moisture within the crop root zone is essential for maximizing yield in precision farming. To monitor soil moisture in real time, modern agricultural IoT stations integrate LoRa radios with soil moisture sensors [11]. Typically, the LoRa radio is installed above ground while the soil moisture sensors are buried underground. However, the above-ground radio components can obstruct routine agricultural operations such as mowing, fertilizing, harvesting, and irrigation [2]. To address this issue, researchers such as Chang et al. [12], and projects such as SoilCare [13] have proposed directly burying LoRa radios underground and estimating soil moisture by analyzing the received LoRa signals from above-ground gateways. This approach enables continuous monitoring without interfering with other agricultural activities.

However, existing works [12], [13] fall short in two aspects. First, they use the phase difference between two antennas to infer the soil moisture. The commercial-off-the-shelf (COTS) LoRa gateways, however, do not provide any interface to obtain such a phase difference. They require a customized software-defined radio to capture raw signals, which incurs additional hardware costs, thereby hampering large-scale deployment. Second, they assume that soil moisture is evenly distributed in soils. However, soil moisture changes over depth due to the dynamic process of water penetration and evapotranspiration in soils [11], [14]. The detection of soil moisture in multiple layers at different depths is more meaningful for precise agriculture. Although CoMEt [14] enables multi-layer soil moisture estimation using wireless signals, it requires custom hardware with a limited communication range compared to LoRa. Thus, detecting multilayer soil moisture with COTS LoRa devices remains an open problem.

To bridge the gap, this paper introduces ISACSoil, an integrated sensing and communication (ISAC) approach that enables multi-layer soil moisture sensing via LoRa cross-soil communication. We notice that the received signal strength indicator (RSSI) is widely available on various COTS LoRa gateways. Moreover, soil moisture changes the path loss of cross-soil signals transmitted by a LoRa node buried in the soil. The basic idea of ISACSoil is to develop a fine-grained signal-path loss model that accounts for signal attenuation and propagation direction to infer moisture levels in multi-layer soils from the transmission power of a LoRa node, the RSSI at a LoRa gateway, and the relative positions of the transceivers. To develop the cross-soil model, we face two challenges as follows:

First, given the soils in a field, we need to know how soil moisture affects the RSSI of cross-soil signals. We know that soil water contributes to signal attenuation. The higher the soil moisture, the greater the signal attenuation. However, there is no theoretical model that quantifies this relationship across diverse soil types. In addition to signal accumulation, complex soil components lead to complex signal reflection, refraction, and scattering, degrading signal strength, including polarization misalignment [2]. Thus, if we aim to build an empirical model using a site survey, it is challenging to extract the signal attenuation under varying soil-moisture conditions during cross-soil propagation.

Second, estimating the propagation direction and path distance is critical to calculating the signal path loss. However, in our multilayer soil moisture sensing, soil moisture levels vary across soil layers. Differences in water content between

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adjacent layers will cause signal reflection and refraction at the boundary. Reflection reduces signal power. Meanwhile, refraction changes the signal's propagation path. Moreover, we do not know the distribution of soil moisture in these layers in advance. It is challenging to model the power loss and propagation changes induced by soil moisture heterogeneity.

To address the two challenges, we first develop a method to filter out polarization misalignment based on the existing cross-soil LoRa radio [2]. We then derive an empirical model relating in-soil signal attenuation to soil moisture. Specifically, the cross-soil radio allows us to adjust the antenna polarization. Thus, the maximum RSSI, with the polarization aligned, reflects the signal attenuation. We develop a heuristic method to obtain the maximum RSSI, rather than scanning all possible polarization settings. According to signal attenuation theory, soil moisture is expected to influence signal attenuation. Given a field, we measure the maximum RSSI to construct an empirical model that quantifies the influence. (Section III-C)

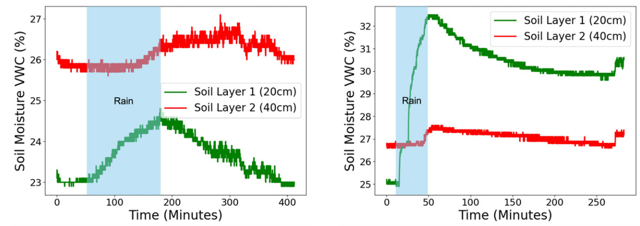
Second, based on the characteristics of electromagnetic wave propagation [15], we find that the multi-layer cross-soil propagation path is unique and can be modeled as a set of separated segments within the soil layers, with ordered soil moisture levels. Specifically, in each layer, the signal propagates along a segment, experiencing attenuation that depends on both soil moisture and the segment length. At the boundary between two adjacent layers or the top layer of soil and the air, the signal is reflected and refracted. Reflection degrades the signal, and refraction changes the propagation direction. We build the relationships between soil moisture and reflection and refraction. (Section III-D)

Third, with the in-soil signal attenuation model, the position of LoRa transceivers, and the reflection/refraction model, we formulate the multi-layer soil moisture estimation as a searching problem that finds a set of soil moisture values and signal incident angles to ensure that the accumulated signal path loss and propagation path correspond to the RSSI and relative position of the transceiver. We design two steps to solve the optimization problem. First, given estimates of soil moisture values for all soil layers, we develop a ray-tracing method to determine the optimal incident angles that minimize the location error. Sitting upon this step, the second step traverses all possible combinations of soil moisture across layers to minimize the RSSI difference. (Section III-E)

We implement ISACSoil with COTS LoRa hardware, including a COTS Dragino LPS8 gateway [16] and a cross-soil communication LoRa node [2]. We conduct extensive experiments to evaluate the performance in a real soybean farm. The results show that the mean soil moisture error is 1.69% for two-layer soil moisture sensing, representing an over 40% improvement compared to state-of-the-art LoRa-based soil moisture sensing methods.

Our contributions are listed as follows:

- To the best of our knowledge, this is the first work to achieve COST LoRa-enabled long-range and multi-layer soil moisture sensing by integrating a state-of-the-art LoRa cross-soil communication technique.



(a) Moderate Rain

(b) Heavy Rain

Fig. 1. The soil moisture records at two layers (20 cm and 40 cm) under different weather conditions.

- We develop a fine-grained signal-propagation model that accounts for in-soil attenuation, reflection, and refraction. Then, we formalize the multi-layer soil moisture sensing problem as an optimization problem subject to three constraints: relative permittivity, transmitter location, and RSSI value. We propose a two-step searching method to solve the problem of finding the optimal soil moisture.
- We implement ISACSoil with COTS LoRa hardware and deploy the system in a real soybean field to evaluate its efficiency. The evaluation results show that the ISACSoil VWC estimation error is approximately 1.69%, representing a 40% reduction compared to the baselines.

II. Background and Preliminary

A. Multi-layer Soil Moisture

We conduct empirical experiments to illustrate the multi-layer soil moisture estimation problem. During the growing season in a soybean field, we deploy two PINO-TECH Soil Watch 10 sensors [17] at depths of 20 cm and 40 cm. We demonstrate the soil moisture difference observed by the two soil moisture sensors under two rainy days with different amounts of water. As shown in Figure 1, the green lines and red lines indicate the soil moisture trend at the 20 cm soil layer and the 40 cm soil layer, respectively. The higher the VWC value, the wetter the soils. We clearly observe the difference in soil moisture. Figure 1(a) shows an approximately 3% VWC difference at most. Figure 1(b) shows an approximately 5% VWC difference at most. Moreover, we observe that rain influences soil moisture across layers in distinct ways. In the shallow layer (i.e., 20 cm, green lines), the rain induces greater changes than in the deep layer (i.e., 40 cm, red lines). The reason is that not all rainwater can penetrate deep into the soil before it evaporates. In addition, the changes in the deep layer occur later than those in the shallow layer. Finally, we observe that deep-layer soils are usually wetter than shallow-layer soils, even under moderate rainfall, as shown in Figure 1(a). However, short-term heavy rain (e.g., thunderstorm) will make the soil moisture in the shallow layer higher than the deep layer as shown in Figure 1(b). Thus, it is meaningful to estimate soil moisture at multiple layers simultaneously.

B. LoRa Cross-Soil Communication

LoRa employs Chirp Spread Spectrum (CSS) modulation, enabling long-range communication with low power usage

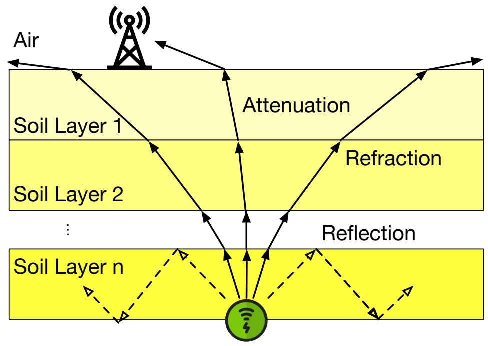


Fig. 2. The overview of ISACSoil soil and signal propagation model: n soil layers, in-layer attenuation, cross-layer reflection, and refraction.

and inexpensive devices [1], [18]. Signal attenuation and polarization misalignment jointly contribute to the loss of cross-soil signals. Since signal attenuation during soil propagation is fundamentally constrained by physical laws and difficult to mitigate, state-of-the-art cross-soil communication designs, such as Demeter [2], have focused on addressing polarization misalignment instead. It proposed a low-cost, low-power programmable antenna design to keep reliable cross-soil communication automatically. Specifically, a LoRa node equipped with a Demeter antenna, called a Demeter node, can configure its polarization by adjusting the phase-shifter settings on the antenna. Thus, it can achieve polarization alignment by searching for a phase shifter setting that maximizes the SNR of the received signals.

A Demeter node can mitigate the loss of signal power due to polarization misalignment. The left signal-path loss, including attenuation and reflection, is related to soil moisture. This motivates us to use the COTS Demeter node and LoRa gateway to enable long-range, multi-layer soil moisture estimation. Next, we conduct an empirical study to illustrate the differences in soil moisture across soil layers and the signal power loss in cross-soil communication.

III. System Design

A. Soil Model and Problem Overview

As shown in Figure 2, we define several components to formalize the problem of multi-layer soil moisture sensing. First, we adopt the soil model in CoMEt [14], which partitions the soil above a LoRa Demeter node into n layers. Different layers have different soil moisture levels. Several factors, including rainfall, snow, irrigation, and evaporation, affect soil moisture. For example, during rainy days, water can take hours to penetrate the soil layers from the top down. On sunny days, the top soil layers dry out faster than the lower layers. Due to the dynamic nature of soil moisture, the distribution of soil moisture levels across these layers can be complex. To simplify our analysis, we adopt an assumption similar to that in existing multi-layer soil moisture sensing [14]: the soil moisture in each layer is uniform, and the depth of each layer is the same. Although the soil model is more precise for small depths, its complexity increases with the number of layers. We can control the depth to trade off the precision and model complexity.

Second, the signal path loss primarily arises from two components: in-layer attenuation and cross-layer reflection. As shown in the solid arrows, the in-layer attenuation indicates the signal attenuation when it traverses a soil layer. As the dashed arrows indicate, cross-layer reflection occurs when portions of the signal are reflected back due to soil-moisture heterogeneity. We can ignore the influence of in-soil multi-path effects due to reflection. The reason is that in-soil attenuation is much greater than that in the air, making multi-path signals very low-power compared to the direct path emanating from the soil.

Moreover, the propagation path between the in-soil LoRa node and the above-ground LoRa gateway is segmented due to cross-layer refraction, indicating a change in propagation direction when the signal crosses the boundary between two adjacent soil layers or between the top soil layer and the air. Because the permittivity of wet soils is much higher than that of air, particularly during the growing season, LoRa signals tend to propagate away from the direction perpendicular to the boundary between the topsoil and the air. This means that the LoRa signal can escape only from a small area of the top-layer soil, making the direct path unique.

Next, we illustrate our design for modeling soil conductivity, extracting in-layer attenuation, and quantifying cross-layer reflection and refraction. Finally, we present our multi-layer soil moisture estimation strategy and its optimization process.

B. Soil Permitivity Model

The soil relative permittivity ϵ_S determines the influence of soils on the LoRa signal propagation. ϵ_S is a complex number, which can be expressed as:

$$\sqrt{\epsilon_S} = \alpha - \beta j \tag{1}$$

α and β represent the real and imaginary parts of the soil relative permittivity. Soil moisture, measured by Volumetric Water Content (VWC) [2], [14], is a major factor in determining ϵ_S. VWC denotes the ratio of the water volume to the total volume of water and soil. Next, we present the models used to compute α and β for a given VWC level.

C. In-soil Attenuation Model and Noise Removal

When wireless signals propagate in a free space, the channel $h(f,d)$ is illustrated by:

$$h(f,d) = \frac{A}{d} e^{-j2\pi f \frac{d}{c}} \tag{2}$$

where A is the attenuation constant determined by antenna property, f is the frequency, d is the distance of the propagation, and c is the light speed. When wireless signals are transmitted through soils with identical soil moisture, their speed is reduced by water between soil particles. Given the soil relative permittivity ϵ_s, the in-soil channel h_{soil} is given by:

$$h_{soil} = \frac{A}{d} e^{-j2\pi f \frac{d\sqrt{\epsilon_s}}{c}} \tag{3}$$

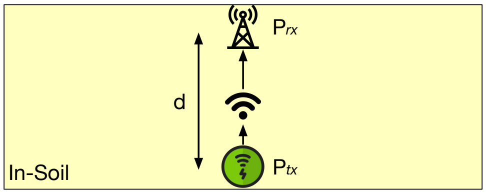


Fig. 3. The illustration of the system settings to measure the in-soil signal attenuation.

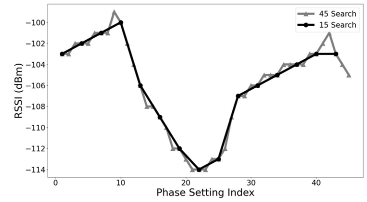


Fig. 4. The RSSI changes under different configurations.

With Equation 1, Equation 3 can be converted to:

$$h_{soil} = \frac{A}{d}e^{-j2\pi f\frac{d\sqrt{\epsilon_S}}{c}} = (\frac{A}{d}e^{-2\pi f\frac{d\beta}{c}})e^{-j2\pi f\frac{d\alpha}{c}} \quad (4)$$

We can see that α influences the speed of phase change by lowering the propagation speed of the wireless signal. Existing works of using the phase change to estimate soil moisture [12], [14], [19] utilize the Topp Equation [20] to mapping the relationship between α and VWC as:

$$\begin{aligned} VWC =& 4.3\times10^{-6}\alpha^3 - 5.5\times10^{-4}\alpha^2 + 2.92\times10^{-2}\alpha \\ & - 5.3\times10^{-2} \end{aligned} \quad (5)$$

Now, we can calculate α under a VWC level with the equation above. However, it is unclear how to calculate β from the VWC. Based on Equation 4, β changes the amplitude, leading to an increase in signal attenuation. Next, we use RSSI from cross-soil LoRa communication to empirically model the relationship between β and VWC levels.

As shown in Figure 3, we bury a LoRa Demeter node and a COTS LoRa gateway in the soil with distance d to establish the in-soil attenuation model. We adjust d to ensure that the soil moisture is kept identical in the soil layer. The transmission power of the LoRa Demeter node is P_{tx}. Then, given the soil channel illustrated in Equation 4, after the in-soil propagation, the received power P_{rx} is shown as follows:

$$P_{rx} = |h_{soil}|^2 P_{tx} = (\frac{A}{d}e^{-2\pi f\frac{d\beta}{c}})^2 P_{tx} \quad (6)$$

With the received power, we can calculate the attenuation-only RSSI, $RSSI_A$, as:

$$\begin{aligned} RSSI_A &= 10\log_{10}(1000P_{rx}) \\ &= 10\log_{10}(1000\frac{A^2}{d^2}P_{tx}) + 10\log_{10}(e^{-4\pi f\frac{d\beta}{c}}) \\ &= 10\log_{10}(1000\frac{A^2}{d^2}P_{tx}) - \frac{40}{\log_e 10}\pi f\frac{d}{c}\beta \\ &= RSSI_{air} - \frac{40}{\log_e 10}\pi f\frac{d}{c}\beta \end{aligned} \quad (7)$$

$RSSI_{air}$ indicates the RSSI in free space given the same transceiver settings, which is a constant value and irrelevant to β. Thus, while keeping the distance d between the transceiver unchanged, the in-soil attenuation is linearly changed with β.

In addition to the theoretical analysis, we empirically study the relationship between RSSI and VWC in a real field with the same settings. Specifically, we manually change the VWC of the soils and record the RSSI received at the COTS LoRa gateway. However, as shown in Section II, the raw RSSI includes both in-soil attenuation and power loss incurred by polarization misalignment. To mitigate the RSSI degradation caused by polarization misalignment and retain only the in-soil attenuation, we need to identify the polarization-aligned configuration of the Demeter node, in which we observe the highest RSSI among other settings.

We can traverse all 45 polarization configurations of a Demeter LoRa node to search for the optimal one. The grey curve in Figure 4 shows an example of the polarization search with all 45 polarization configurations. We observe that the RSSI follows a sine-wave distribution across all configurations. Moreover, the RSSI values near the maximum are close to it. Thus, instead of scanning all polarization configurations, we uniformly scan 15 configurations, as the black curve shows in Figure 4, to balance the energy efficiency for long-term experiments and the accuracy of the obtained maximum RSSI. Given a series of tuples [maximum RSSI, VWC], where each tuple indicates the maximum RSSI for a given VWC, we use regression to fit an empirical equation relating attenuation-only RSSI to VWC. Together with Equation 7, we can calculate the imaginary part of the soil permittivity, β.

D. *Modeling Signal Reflection and Refraction*

When LoRa signals cross the boundary between two adjacent soil layers or between the top soil and the air, portions of the signals are reflected back. According to the theory of electric-magnetic wave propagation [15], the power loss can be indicated as follows:

$$\frac{P_r}{P_t} = |\frac{\sqrt{\epsilon_1} - \sqrt{\epsilon_2}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}}|^2 \quad (8)$$

P_r is the reflected power, and P_t is the incident power. The wireless signals propagate from a material with relative permittivity ϵ_1 to another material with relative permittivity ϵ_2.

In addition to the reflected signals, the other part of the LoRa signals traverses the boundary and enters another soil layer or the air. Due to the different material permittivity in the new layer, the refraction will change the propagation direction. According to the theory of electric-magnetic wave propagation [15], the propagation direction change can be illustrated as follows:

$$\alpha_1 \sin\theta_1 = \alpha_2 \sin\theta_2 \quad (9)$$

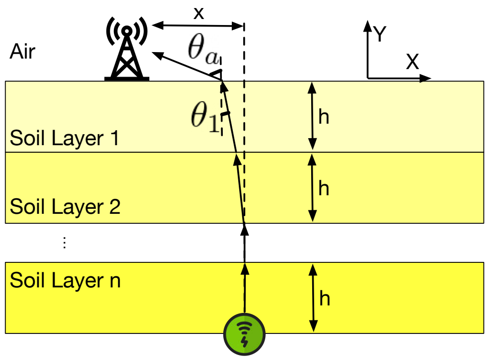


Fig. 5. Path Loss Model: The signal traverses multiple soil layers to reach the LoRa gateway with attenuation, reflection, and refraction.

θ_1 is the incidence angle, and θ_2 is the refraction angle. Corresponding to Equation 1, α_1 and α_2 are the real parts of the relative permittivities of the two materials $\sqrt{\epsilon_1}$ and $\sqrt{\epsilon_2}$.

E. Soil Moisture Estimation Model

Based on the models established above, we formulate our soil moisture estimation problem as an optimization problem. As shown in Figure 5, we have n soil layers. For the soil layer i, the incidence angle is θ_i, and its VWC is κ_i. The incidence angle in the air is θ_a. The depth of each soil layer is h. The relative permittivity of each soil layer is $\sqrt{\epsilon_i}$, and its real part and imaginary part are indicated as α_i and β_i, respectively. The height of a LoRa gateway is h_g. The horizontal distance between the LoRa gateway and the transmitter is x. The transmitted power of the transmitter is P_{tx}. The received RSSI is ϕ. Then, we have the following three constraints.

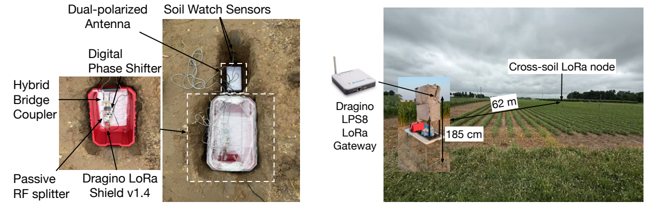
The first constraint is the relative permittivity constraint, which indicates that the relative permittivity $\sqrt{\epsilon_i}$ and soil moisture κ_i of each soil layer should comply with the relationship illustrated in Sections III-B and III-C. Specifically, given a soil moisture VWC κ_i in the i^{th} soil layer, we can calculate the corresponding α_i with Equation 5 and infer β_i with the empirical method in Section III-C.

The second constraint is the location constraint, which indicates that the propagation direction has to fulfill the geometric distance, and the relationship among incident angles fulfills Equation 9. We illustrate the detailed constraints as follows:

1) $\sum_{i=1}^{n} \tan\theta_i \cdot h + \tan\theta_a \cdot h_g = x$
2) $\alpha_i \sin\theta_i = \alpha_{i-1} \sin\theta_{i-1}$ for $i \in [2, n]$
3) $\alpha_1 \sin\theta_1 = \sin\theta_a$

The third constraint is the RSSI and path loss constraint, which indicates that the observed maximum RSSI with polarization alignment should correspond to the accumulated signal power loss, including reflection loss and attenuation loss, along the path. For the i^{th} layer, we use P_{in}^i and P_{out}^i to indicate the power of the signal coming in and escaping out of the layer. Then, the RSSI and path constraint can be illustrated as follows:

4) $P_{in}^n = P_{tx}$
5) $P_{out}^i = (\frac{A}{d} e^{-2\pi f \frac{(h/\cos\theta_i)\beta_i}{c}})^2 P_{in}^i$



(a) In Soil LoRa Node (b) LoRa Gateway

Fig. 6. The implementation and deployment of ISACSoil.

6) $P_{in}^{i-1} = (1 - |\frac{(\alpha_i - j\beta_i) - (\alpha_{i-1} - j\beta_{i-1})}{(\alpha_i - j\beta_i) + (\alpha_{i-1} - j\beta_{i-1})}|^2) \cdot P_{out}^i$ for $i \in [2, n]$
7) $P_{in}^{air} = (1 - |\frac{(\alpha_1 - j\beta_1) - 1}{(\alpha_1 - j\beta_1) + 1}|^2) P_{out}^1$
8) $\phi = 10 \log_{10}(1000 \frac{A^2}{(h_g/\cos\theta_a)^2} P_{in}^{air})$

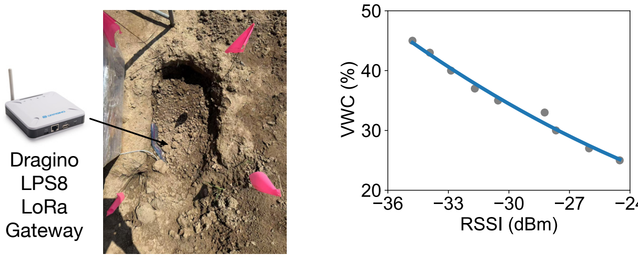
In our optimization problem, n soil moisture VWC values $\{\kappa_1, \kappa_2, ..., \kappa_n\}$ are unknown. According to the relative permittivity constraint, α_i and β_i can be calculated by the soil moisture κ_i. According to the location constraint, given θ_n and the α_i of all n soil layers, we can determine θ_{n-1}, θ_{n-2}, ... , θ_1, and θ_a, respectively. Thus, in these constraints, the independent unknown parameters are n soil moisture levels κ_i and the original incident angle θ_n. In practice, we will notice that the horizontal distance x and RSSI ϕ calculated from these constraints are different from those observed ones $\hat{x}$ and $\hat{\phi}$. Thus, the optimization problem is to search θ_n and $\{\kappa_1, \kappa_2, ..., \kappa_n\}$ to minimize the differences $||x - \hat{x}||$ and $||\phi - \hat{\phi}||$ as follows:

$$\overline{\theta_n}, \overline{\kappa_i}, i \in [1, n] = \arg \min_{\theta_n, \kappa_i, i \in [1,n]} ||x - \hat{x}||^2 + ||\phi - \hat{\phi}||^2 \quad (10)$$

We have two steps to solve the optimization problem. First, we can use ray tracing to solve the optimization problem with respect to $||x - \hat{x}||$. Specifically, given a determinstic set of $\{\kappa_1, \kappa_2, ..., \kappa_n\}$ indexed as j, we calculate the relative permittivity α_i for each soil layer. Then, we search all possible θ_n to find the $\overline{\theta_n}_j$ that minimizes the distance error $||x_j - \hat{x}||$ given equations 1)-3) in the location constraint. Second, we traverse all possible $\{\kappa_1, \kappa_2, ..., \kappa_n\}$. For the j combination of soil moisture values, we execute the first step to determine $\overline{\theta_n}_j$ and calculate $\{\overline{\theta_1}_j, \overline{\theta_2}_j, ..., \overline{\theta_{n-1}}_j, \overline{\theta_a}_j\}$ correspondingly. Then, under such a combination, we use equations 4)-8) in the RSSI and path loss constraint to calculate $||\phi - \hat{\phi}||$. The combination of achieving a minimum $||\phi - \hat{\phi}||^2$ is the final soil moisture estimation for the n soil layers.

IV. IMPLEMENTATION

Figure 6 shows the implementation and deployment. Firstly, as shown in Figure 6(b), we use a COTS Dragino LPS8 gateway [16], deployed at a height of 1.85 m, to receive LoRa signals. The LoRa gateway is powered by a mobile battery. The LoRa gateway and mobile battery are well-protected against high temperatures and rain. Moreover, as shown in Figure 6(a), we use the same way stated in Demeter [2], including a Dragino LoRa Shield v1.4 [21], a 3 dB XRDS-RF 698–2700 MHz passive RF splitter [22], a PE44820 digital phase shifter with 45 discrete phase offset levels [23], a hybrid



(a) Depoyment (b) Data Regression

Fig. 7. The deployment (a) and data regression (b) of the in-soil attenuation calibration.

bridge coupler (FR4), and an ANT627-NF-PANEL-MIMO-OD dual-polarized antenna [24], to implement a Demeter node. The antenna is buried at a depth of 40 cm in the ground. The rest of the LoRa Demeter node is buried beside it in a plastic box to ensure waterproofing. The transmission power is set to 14 dBm. The channel frequency is 904.1 MHz.

In our implementation, each soil layer is 20 cm deep. As shown in Figure 6(a), we deploy two PINO-TECH Soil Watch 10 sensors [17] to monitor soil moisture at two different soil layers, which are 20 cm and 40 cm depth. The soil moisture sensors are connected to the LoRa node. Correspondingly, the soil moisture data are transmitted via LoRa packets, serving as the ground truth. Finally, as shown in Figure 6(b), we deploy our system in a real soybean field during the growing season. The LoRa gateway and the cross-soil node are located on opposite sides of the field. The horizontal distance between the LoRa node and the LoRa gateway is 62 m.

Under these deployment settings, we collected data for 7 days to ensure that soil moisture varies as widely as possible across a range of summer weather conditions. The LoRa Demeter node transmits data packets across 16 uniformly distributed polarization directions for all spreading factor (SF) settings from 7 to 10 every 30 minutes. The soil moisture sensor readings are part of the payload. In the next 8 days, we will adjust the LoRa gateway's location or the buried depth of the LoRa Demeter node to different depths to collect more data with diverse settings.

A. *In-soil Attenuation Calibration*

As illustrated in Section III-C, we need to empirically calibrate the relationship between soil moisture and RSSI so that we can infer the imaginary part of soil relative permittivity, β, with soil moisture VWC. To achieve this calibration, as shown in Figure 7(a), we bury the LoRa gateway, sealed in a plastic bag but leaving the antenna out, in the soil and keep the distance between the transceivers at 20 cm, the same as the depth of a soil layer. Then, we record the maximum RSSI across soil moisture conditions ranging from 25% VWC to 45% VWC. The dots in Figure 7(b) show the records correspondingly. We use the polynomial regression method to obtain the empirical equation between VWC and RSSI as follows:

TABLE I
THE SUMMARY OF SYSTEM PARAMETERS.

Parameter	Symbol	Value
Depth of a soil layer	h	20 cm
Height of the LoRa gateway	h_g	185 cm
Transmission power	P_{tx}	14 dBm
Channel frequency	f	904.1 MHz
Horizontal distance between transceiver	x	62 m
Light speed	c	3×10^8 m/s
Antenna attenuation constant	A	0.0399
Signal attenuation in air	$RSSI_{air}$	0 dB

$$\mathrm{VWC} = 0.043 \cdot \mathrm{RSSI}^2 + 0.6637 \cdot \mathrm{RSSI} + 15.4367 \qquad (11)$$

Moreover, we measure the RSSI in the air while keeping the same distance (i.e., 20 cm) between the transceivers and avoiding above-ground multipath effects by placing both devices on high points. The measured RSSI in the air, $RSSI_{air}$, is 0 dBm. Moreover, with Equation 7, the antenna attenuation constant, A, can be calculated as 0.0399. The search range of the incident angle θ_n and soil moisture VWC is $[0.1°, 30°]$ and [20%, 50%], respectively. The search step is 0.1° and 0.1%. Overall, the implementation parameters of ISACSoil are summarized in Table I.

V. EVALUATION

We use the following data-processing and evaluation metrics to evaluate ISACSoil.

Data Processing: From the collected data traces, in each transmission period, given a deterministic SF, we find the maximum RSSI by searching all 16 polarization directions. In addition, we average the VWC records of the two soil layers in the 16 packets as the ground truth.

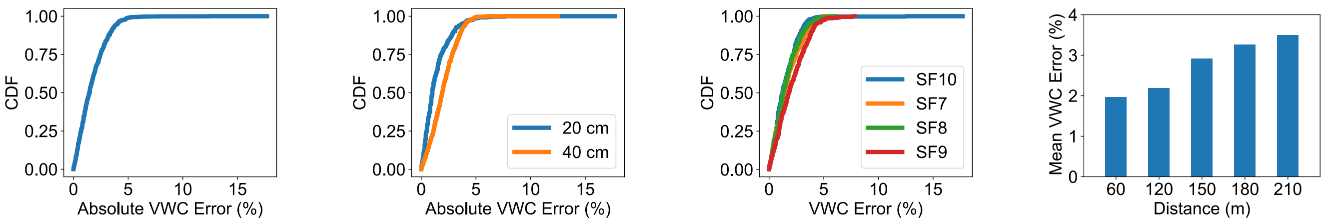
Metrics: We use **VWC error**, defined as the absolute difference between the estimated soil moisture from ISACSoil and the ground truth. It depicts the accuracy of ISACSoil.

A. *Overall Two-Layer Performance*

We examine the overall performance of the two-layer experiments. Figure 8(a) shows the cumulative distribution function (CDF) of all VWC errors. We observe that the mean, 10%, and 90% of VWC errors are 1.69%, 0.31%, and 2.73%, respectively. Compared to the mean VWC error 2.76%–3.1% reported in existing LoRa-based methods [12], [19], ISACSoil improves 38.8% and 45.5% correspondingly. The improvement comes from the fine-grained multi-layer soil and signal propagation model.

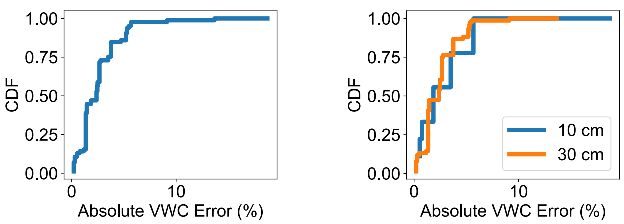
B. *Performance under Different Scenarios*

Figure 8 presents the error of our VWC estimation method under three different settings: different layers, different LoRa communication configurations, and different distances between the cross-soil LoRa node and LoRa gateway. Figure 8(b) shows the CDF of VWC errors at two different soil layers – 20 cm and 40 cm. We observe that the mean errors for the 20 cm and 40 cm layers are 1.34% and 2.04%, respectively.



(a) Overall Performance (b) Different Layers (c) Different SFs (d) Different Gateway Distance

Fig. 8. The detailed performance of ISACSoil under the two-layer experiments (20 cm and 40 cm).



(a) Overall Performance (b) Different Depth

Fig. 9. The performance of ISACSoil in singal-layer experiments.

The deep-layer estimation is slightly less accurate than the shallow-layer estimation. This may be because the soil moisture in the shallow layer is more uniform than in the deep layer. In deep soil layers, a single soil moisture value cannot capture microscale variations, which may lead to unpredictable signal changes.

Moreover, Figure 8(c) isolates the effect of LoRa SF on 20 cm depth measurements. We observe that intermediate SFs deliver the best trade-off between link budget and time-on-air: SF8 achieves the lowest error 1.19%, closely followed by SF7 1.20%, whereas higher SFs, SF9: 1.43%, SF10: 1.55%, suffer increased exposure to channel noise. These results guide practitioners to favor SF7–SF8 for reliable soil moisture sensing.

Finally, Figure 8(d) examines how horizontal transmitter–receiver distance impacts error at 20 cm depth. The VWC error reaches a minimum of 1.97% at 60 m, then rises sharply beyond 120 m, peaking at 3.49% at 210 m. This increasing trend reveals an optimal operating range of 60–100 m for maintaining an error below 2.5% and cautions against excessive link distances, where in-soil attenuation and in-air multipath distortion from crops dominate.

C. Single Soil Layer Experiments

In addition to the multi-layer soil moisture sensing, ISACSoil can be extended to estimate single-layer soil moisture. In our single-layer experiments, we set the soil depth to 10 cm and 30 cm. They differ from the 20 cm setting used in the previous two-layer experiments. In this way, we also verify the influence of layer depth on estimation accuracy.

As shown in Figure 9, the mean error is 4.78% over all experiment results, with the 10 cm depth exhibiting a higher error of 5.32% and the 30 cm depth showing a markedly lower error of 2.55%. Moreover, the results exceed the mean VWC error of 1.69% when the layer depth is set to 20 cm. The elevated error at 10 cm can be attributed to increased near-surface heterogeneity—such as root distribution, surface moisture fluctuations, and soil structure variability—that amplifies signal noise and estimation bias. Conversely, at a depth of 30 cm, the soil moisture profile is more uniform and less affected by surface dynamics, resulting in substantially improved VWC inference accuracy. However, it exhibits greater soil diversity across 30 cm soils than across 20 cm depth soils. Overall, the single-layer performance of 4.78% reflects a trade-off between the high variability near the surface and the stable sensing conditions at greater depth. A depth of 20 cm strikes a good balance given these factors.

VI. Related Work

LoRa-based Soil Moisture Sensing. Smol [25] uses commodity LoRa devices to infer soil moisture from signal variations, but ignores polarization misalignment, leading to large errors and limiting it to surface-level sensing. Chang et al. [12] and SoilCare [19] leverage LoRa phase and deep learning, but rely on SDR platforms, hindering practical deployment. In contrast, ISACSoil enables fine-grained multi-layer sensing using COTS LoRa devices and cross-soil communication.

Other RF-based Soil Moisture Sensing. LTE [26] and Wi-Fi [27] based approaches estimate moisture from phase changes, but require SDR hardware and are limited to single-layer sensing, with shorter communication ranges. CoMEt [14] supports multilayer sensing but depends on precise antenna alignment and SDR platforms, and operates at higher frequencies with limited range. In contrast, ISACSoil enables practical multi-layer soil moisture estimation over long distances with COTS hardware.

Battery-less Soil Moisture Sensing. RFID-based methods [13] and chipless tags such as SoilTag [28] and SoilNutri [29] estimate moisture via signal reflections, but suffer from short range and sensitivity to deployment. UWB radar [30] and MetaSoil [31] enable remote sensing but require complex infrastructure and custom hardware. These approaches lack scalability and do not support practical multilayer sensing.

In contrast, ISACSoil leverages low-cost, low-power COTS LoRa devices for scalable, long-range multilayer soil moisture sensing.

VII. Discussion

Computation Complexity vs. Soil Realism: RSSI measurements are decoded at the LoRa gateway and processed in the cloud, making the computational overhead manageable. The complexity scales with the number of assumed soil layers: more layers increase computation but better capture soil heterogeneity. In practice, the number of layers can be adjusted based on soil type and season to balance computational cost and modeling fidelity.

Robustness to Channel Noise: To balance transmission overhead and RSSI accuracy, we perform partial scans (e.g., 16 packets) to approximate optimal polarization alignment. Residual channel noise may degrade estimation accuracy, with worst-case errors exceeding 10% in our experiments. However, errors remain moderate in typical conditions.

System Scalability: Our system requires an initial calibration using known transceiver positions, burial depth, and the number of soil layers. This process ensures identifiability and solution uniqueness, enabling reliable deployment across different fields and growing seasons.

VIII. Conclusion

We present a LoRa-based sensing system that integrates communication and sensing to estimate soil moisture across multiple layers. Our approach combines an in-soil attenuation model, a calibration method to isolate signal attenuation, and modeling of reflection and refraction across soil interfaces. We formulate multi-layer moisture estimation as a constrained optimization problem and solve it with a two-step method to minimize error. Implemented on COTS LoRa devices and evaluated in a real soybean field, our system achieves a mean VWC error of 1.69% in two-layer scenarios, over 40% lower than existing LoRa-based methods, demonstrating its accuracy and practicality. We plan to evaluate our system in fields with different soil types, crops, and burial depths in our future work.

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